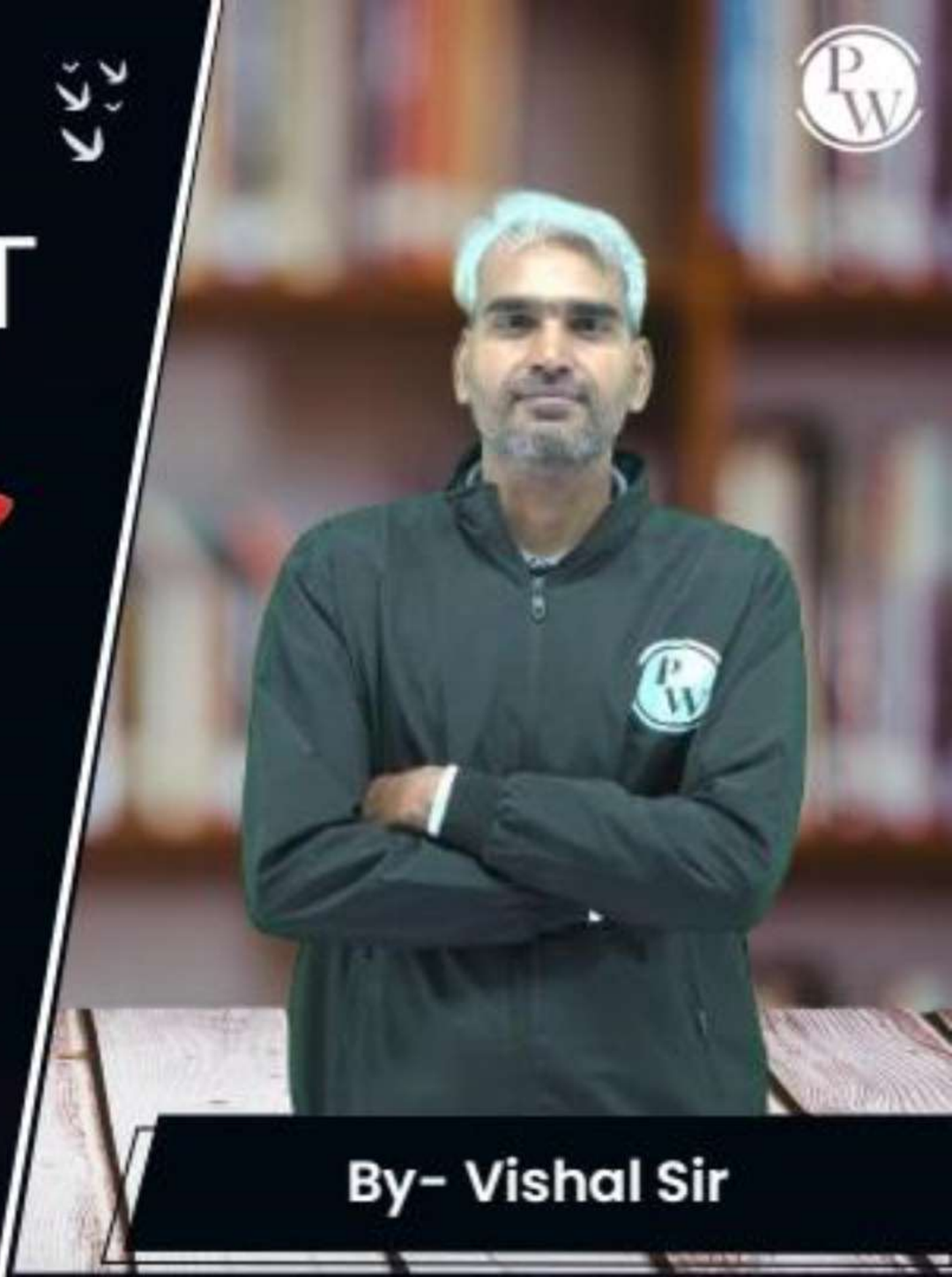
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



Topic

Functional dependency

Slide

Topics to be Covered



- Topic: Functional dependency
- Topic: Types of functional dependency
- Topic: Properties of functional dependency
- Topic: Different types of keys in RDBMS

Slide

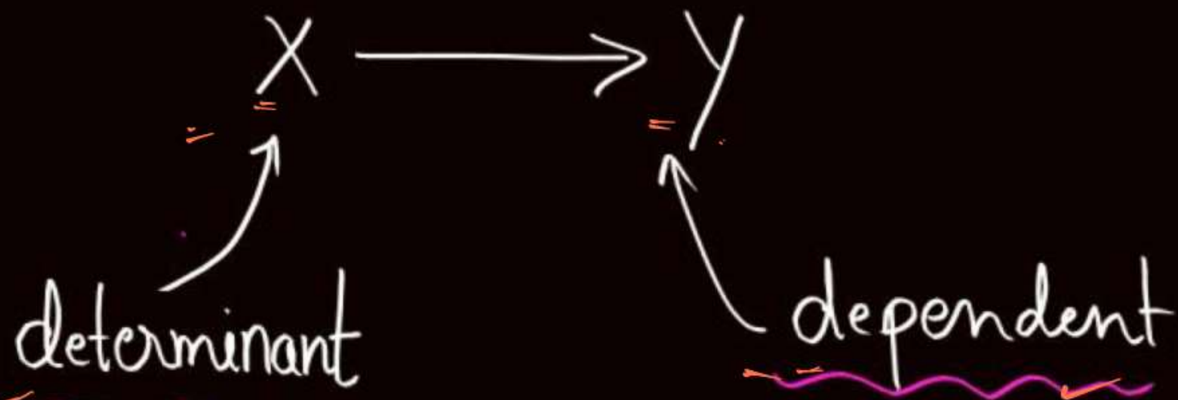


Topic : Functional dependency (FD)

$X \rightarrow Y$

- Functional dependency defines the relationship between two sets of attributes in a relational table.
- It states that the values of attributes of one set (the dependent attribute set) is completely determined by the values of attributes of another set (the determinant attributes). In other words, if a determinant attribute's value is known, the value of the dependent attribute can be deduced.

In FD



* $X \rightarrow Y$ exists in relation R means, whenever values of attributes of set X {determinant set} is known then we can uniquely determine the values of attributes of set Y {dependent set}



Topic : Functional dependency (FD)

P only if Q
≡ if P then Q

Whenever if Condⁿ is true then then Condⁿ will also be true

- Let R be the relational schema with X and Y as the attribute sets over relation R.

Functional dependency X→Y exists in R only if
For all pair of tuples t1,t2∈R
If t1.X = t2.X then t1.Y = t2.Y

X → Y

If functional dependency X→y exists in R,
then,
for pair of tuples $t_1, t_2 \in R$
If $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$

X R Y

Att_1	Att_2	Att_3	Att_4	Att_5	Att_6
a	b	c	p	q	
a	b	c	p	q	

* Let R be a relation, and functional dependency $X \rightarrow Y$ exists in R,

then Whenever Value of X is same in any pair of tuples, then Value of Y should also be same in those tuples

* If $X \rightarrow Y$ exists in R,
  - then whenever values of attribute set X is same, the values of attribute set Y must also be same
  - But if values of attribute set Y are same then value, of attribute set X need not be same

#Q. Consider the following relational instance

A	B	C
1	2	3
1	2	4
2	2	1
3	1	2
[redacted]		

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $A \rightarrow C$

b) $A \rightarrow B$

c) $AB \rightarrow C$

d) $BC \rightarrow A$

#Q. Consider the following relational instance

A	B	C
1	1	1
1	2	2
2	4	3
3	3	4
4	1	5
5	3	6

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $B \not\rightarrow C$

b) $A \not\rightarrow B$

c) $C \rightarrow B$

d) $B \not\rightarrow A$

e) $C \rightarrow A$



Topic : Functional dependency (FD)

whenever X is same values of Y in those tuples should also be same

- If necessary condition for functional dependency "X→Y" does not hold true based on given relation instance, then functional dependency "X→Y" can never exist in the given relation.

- Even if necessary condition for functional dependency "X→Y" does hold true based on given relation instance, then also we can not be sure whether functional dependency X→Y exists in the relation or not, because it is just the relational instance.

"MCQ" - GATE-2002

#Q. From the following instance of a relation schema R (A,B,C), we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

$A \rightarrow B$ May or may not exist.

$B \not\rightarrow C$

We can definitely Conclude, that B does not determine C

(A) A functionally determines B, and B functionally determines C

(B) A functionally determines B, and B does not functionally determine C

(C) B does not functionally determine C

(D) A does not functionally determine B, and B does not functionally determine C

May or May Not



Topic : Types of Functional Dependency

Three Types

- Trivial FD
- Non-trivial FD
- Semi non-trivial FD

Let, Student can enroll only for one branch

eg:

Student

Sid	Sname	Branch
S_1	A	CS
S_2	A	IT
S_3	B	CS

Sname → Sname

Sid, Sname → Sname

example of trivial FD



Topic : Types of Functional Dependency

Three Types

- Trivial FD
- Non-trivial FD
- Semi non-trivial FD

Let, Student can enroll only for one branch

eg:

Student

Sid	Sname	Branch
S_1	A	CS
S_2	A	IT
S_3	B	CS

Sid → Sname

Sid → Branch

Example of non-trivial FD



Topic : Types of Functional Dependency

Three Types

- Trivial FD
- Non-trivial FD
- Semi non-trivial FD

Let, Student can enroll only for one branch

eg:

Student

Sid	Sname	Branch
S_1	A	CS
S_2	A	IT
S_3	B	CS

(Sid, Sname) → (Sname, Branch)

Already know

New info.

eg. of Semi non-trivial FD



Topic : Trivial Functional Dependency

Let R be the relational schema, and X and Y are any two attribute sets over relation R such that $X \supseteq Y$,

then $X \longrightarrow Y$ is called trivial FD.

* All trivial FDs which are possible w.r.t. attribute of given relation will always hold true in that relation.

↑ Trivial FDs are always implied in the Relation



Topic : Non-trivial Functional Dependency

* Let R be the relational schema, and X and Y are the attribute sets over relation R.

* If functional dependency $X \rightarrow Y$ exists in relation R, and $X \cap Y = \phi$, then $X \rightarrow Y$ is called non-trivial FD

eg let $AB \rightarrow BC$

here $AB \cap BC = B \neq \phi$

∴ not a non-trivial FD

eg $AB \rightarrow CD$

$AB \cap CD = \phi$

∴ Non-trivial FD



Topic : Semi non-trivial Functional Dependency

Cond^n for

$X \longrightarrow Y$

to b.

Semi - Non-trivial

* Let R be the relational schema and X and Y are the attribute sets over relation R.

* If functional dependency $X \longrightarrow Y$ exists in relation R, s.t. $X \not\supseteq Y$ and $X \cap Y \neq \emptyset$, then $X \longrightarrow Y$ is called Semi non-trivial FD

Note :-

1. Every trivial FD is always implied in the relation, therefore we don't need to worry about trivial FDs.
2. We can always seperate semi-non-trivial FD into trivial and non-trivial part
3. Our main concern is non-trivial FDs.

#Q. Consider the following relational instance

A	B	C
1	1	1
1	1	2
2	1	2
2	2	3
3	3	4

Possible non-trivial FDs over three attributes A, B and C are,

$A \rightarrow B$ ✗
$A \rightarrow C$ ✗
$B \rightarrow A$ ✗
$B \rightarrow C$ ✗
$C \rightarrow A$ ✗
$C \rightarrow B$ ✓

$A \rightarrow BC$ ✗
$B \rightarrow AC$ ✗
$C \rightarrow AB$ ✗

$AB \rightarrow C$ ✗
$AC \rightarrow B$ ✓
$BC \rightarrow A$ ✗

Find all non-trivial FDs which may hold true in the above relation based on given relational instance.

May exist

FDs which may exist in the above Relⁿ are,
$C \rightarrow B$ & $AC \rightarrow B$

H.W.

#Q. Write all possible non-trivial FDs with respect to relational schema

R(A, B, C, D)

Also try to identify the number of non-trivial FDs Using Combinatorics



Topic : Properties of Functional Dependencies

These three Properties are called Armstrong's Axioms

1. Reflexivity: (Trivial FDs) If $X \supseteq Y$, then $X \rightarrow Y$ is said to satisfy reflexivity

2. Augmentation:- Let $X \rightarrow Y$ exists in relation R and Z is some attribut set from relation R, then $XZ \rightarrow YZ$ holds true

3. Transitivity: if $X \rightarrow Y$ and $Y \rightarrow Z$ exists in Relation R, then $X \rightarrow Z$ holds true



Topic : Properties of Functional Dependencies

④ Decomposition (Splitting Rule) : if $X \longrightarrow YZ$ exists in Rel^n R, then $X \longrightarrow Y$ and $X \longrightarrow Z$

Note : if $X \longrightarrow YZ$, then $X \longrightarrow Y$ & $X \longrightarrow Z$ holds true, but if $XY \longrightarrow Z$, then $X \longrightarrow Z$ & $Y \longrightarrow Z$ need not hold true

* $AB \longrightarrow BC$ :: $AB \longrightarrow B$ } trivial part
& $AB \longrightarrow C$ } Non-trivial part

$AB \longrightarrow BC$: Semi non trivial



Topic : Properties of Functional Dependencies

(5) Union : (Composition) if $X \rightarrow Y$ and $X \rightarrow Z$ exists in R, (LHS is same) then $X \rightarrow YZ$ holds true

also if $X \rightarrow Y$ and $A \rightarrow B$ exists in R
then $XA \rightarrow YB$ holds true



Topic : Different types of keys

H.W

1. Candidate key
2. Primary key
3. Alternate key
4. Super key

Reading



2 mins Summary

Topic Functional dependency

Topic Types of functional dependency

Topic Properties of functional dependency

Topic Different types of keys in RDBMS

Slide

THANK - YOU